

Chapter 1. Introduction: Linear and non-linear models

Neural Networks

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Máster Universitario en Inteligencia Artificial, Reconocimiento
de Formas e Imagen Digital

Departamento de Sistemas Informáticos y Computación

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Neural networks

- **Goal:** Introduce the main techniques based on neural networks that deal with problems related to pattern recognition and machine learning.
- Focus on *automatic learning from training data and recognition*.
- Neural networks are inspired in the human ability to recognize patterns.
- ***A densely interconnected set of elemental processors.***
- Other known names:
 - A connectionist models.
 - Artificial neuronal networks.
 - Distributed and parallel processing.
- Problem: Interpretability (Tools: SHAP, ...)

Adquisition, representation and recognition

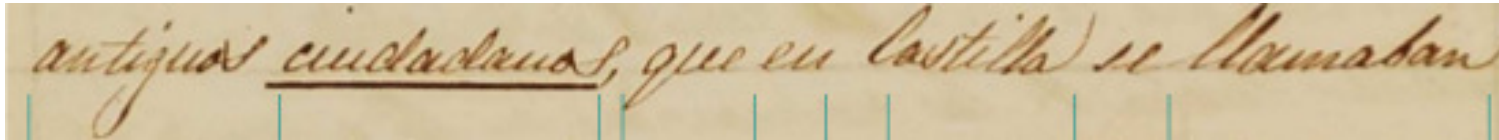
- Perception entails a process of: information acquisition, representation and recognition:



- Sensors: Camera, microphone, ...
- Signals: Image file, data streaming, ...
- Representations (x): Feature vectors, string of symbols, graphs,...
- Interpretation: class, string of symbols, real number, vector, ...

Recognition and interpretation

- Usually, pattern recognition is considered as the process to assign a *class* label to the objects that are properly acquired and represented
- However, pattern recognition can be a more general topic where the result of the process is not a class label but a vector (regression) or something more structured, e.g. a sentence, a graph, ...:



- But in this subject we will refer normally to the *classification* problem

Tools

- TensorFlow (An end-to-end open source machine learning platform)
<https://www.tensorflow.org/>
- CUDA (A development environment for creating high performance GPU-accelerated (NVIDIA) applications)
<https://developer.nvidia.com/cuda-toolkit>
- Keras (running on top of TensorFlow, CNTK, or Theano.)
<https://pypi.python.org/pypi/Keras>
- PyTorch (A scientific computing framework with wide support for machine learning algorithms that puts GPUs first. Previously Torch)
<https://pytorch.org/>
- Caffe (A deep learning framework) <http://caffe.berkeleyvision.org/>

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Classes, representation space and classifiers

- **Objects & classes:** U and $\mathbb{C} = \{1, 2, \dots, C\}$
 - Each *object* (or its signal) is showed in a *Primary Space* or “Universe”, U
 - Let's assume that each object $x \in U$ belongs to a unique *class* $c(x) \in \mathbb{C}$
 - $\mathbb{C}$ is the set of all possible *identifiers* or *class labels*
- **Representation Space:** E , generally $E = \mathbb{R}^D$
 - Let $\mathbf{x} = y(x)$ be the result of the preprocessing and feature extraction process applied to an object $x \in U$
 - E encloses all the possible results: $\{\mathbf{x} : \mathbf{x} = y(x), x \in U\} \subset E$
 - Given that two different objects from U can have the same representation in E , it is not guaranteed that each point in E belongs to a unique class.

Classifiers and discriminant functions

Classifier: $G_{\mathbf{w}} : E \rightarrow \mathbb{C}$

- $G_{\mathbf{w}}$ is learnt with N labelled samples $(\mathbf{x}_1, c_1), \dots, (\mathbf{x}_N, c_N) \in E \times \mathbb{C}$
- The goal is to obtain the correct class for a new object $x \in U$:
$$G_{\mathbf{w}}(y(x)) = G_{\mathbf{w}}(\mathbf{x}) \stackrel{!}{=} c(x)$$

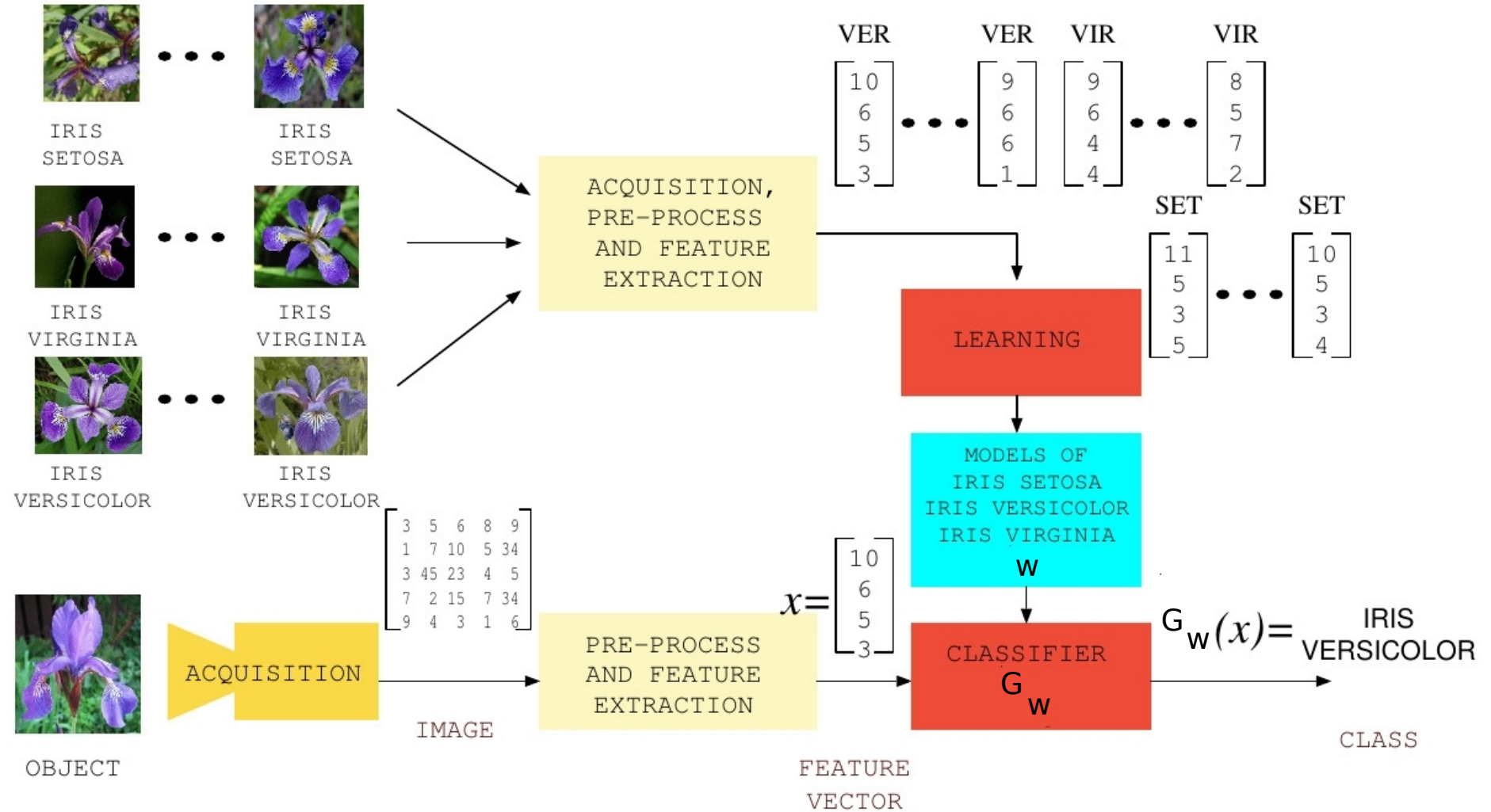
... the maximum number of times as possible (how to reach a 100%?)

Every classifier $G_{\mathbf{w}}$ into C classes can be stated in terms of C **discriminant functions** $g_c : E \rightarrow \mathbb{R}$, $1 \leq c \leq C$, and the corresponding *classification rule*:

$$\hat{c} = G_{\mathbf{w}}(\mathbf{x}) \equiv \operatorname{argmax}_{1 \leq c \leq C} g_c(\mathbf{x}; \mathbf{w})$$

Notation: $G \equiv G_{\mathbf{w}}$ and $g_c(\mathbf{x}) \equiv g_c(\mathbf{x}; \mathbf{w})$

Classes, representation space and classifiers



Decision or classification boundaries

- **Decision regions:** Any classifier partitions the representation space into C decision regions, $R_1, \dots, R_C$:

$$R_j = \{\mathbf{x} \in E : g_j(\mathbf{x}) > g_i(\mathbf{x}) \quad i \neq j, \quad 1 \leq i \leq C\} \text{ for } 1 \leq j \leq C$$

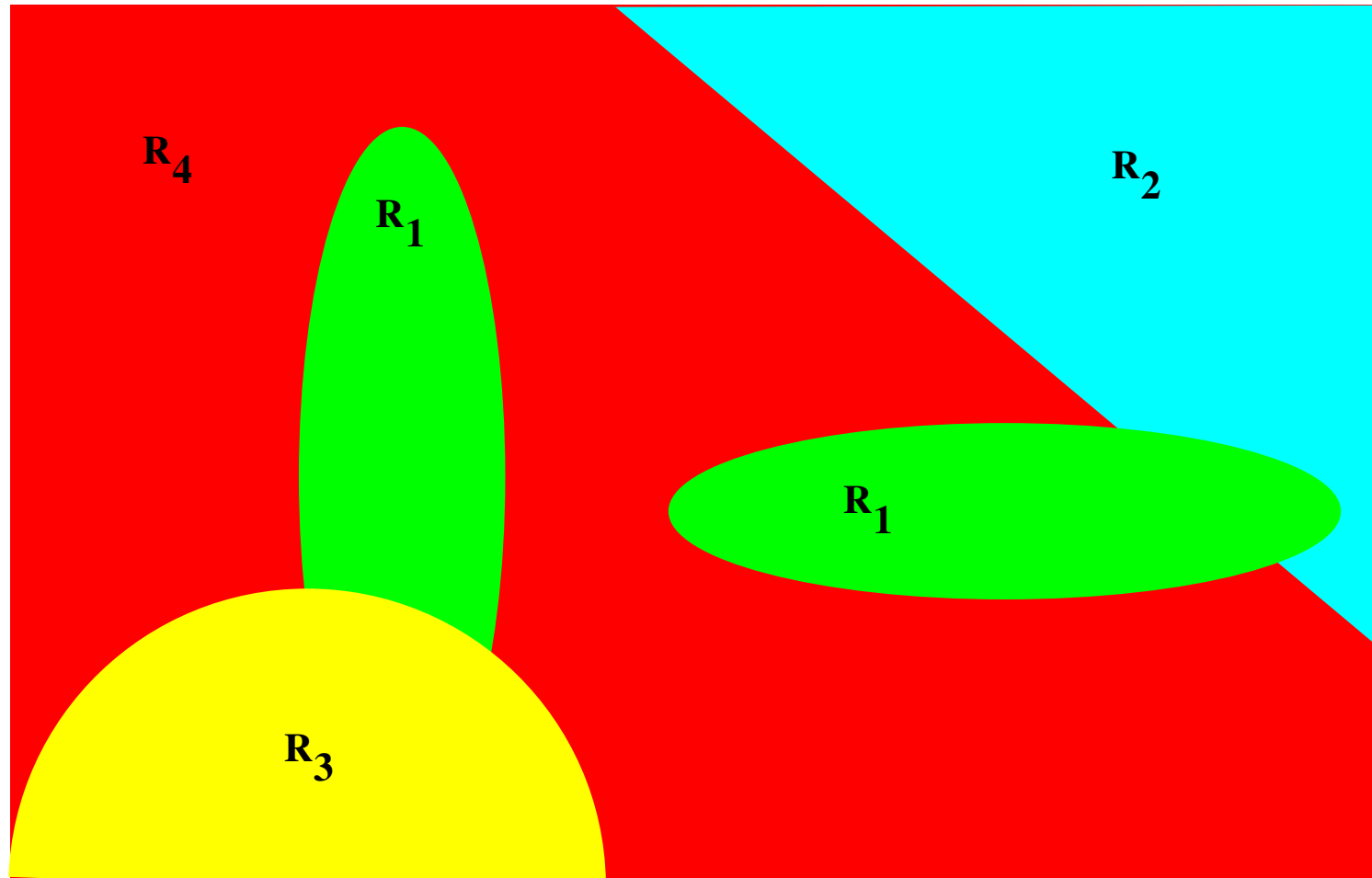
- **Decision boundary between two classes i, j for $1 \leq i, j \leq C$:**
Set of points $\mathbf{x} \in E$ for which $g_i(\mathbf{x}) = g_j(\mathbf{x})$

In general they are *hypersurfaces* defined by the equations:

$$g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0 \quad i \neq j, \quad 1 \leq i, j \leq C$$

- If $E \equiv \mathbb{R}^3$, the boundaries are surfaces (ej. *planes*)
 - If $E \equiv \mathbb{R}^2$, the boundaries are lines (ej. *straight lines*)
 - If $E \equiv \mathbb{R}$, the boundaries are points
- **Decision boundary of a single class i for $1 \leq i \leq C$:**
Set of points $\mathbf{x} \in E$ for which $g_i(\mathbf{x}) = \max_{j \neq i} g_j(\mathbf{x})$

Regions and decision surfaces



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Linear discriminant functions (LDFs)

- A classifier is *linear* if its discriminant functions are *linear functions* of the vectors of E . Let be $\mathbf{x} \in E \equiv \mathbb{R}^D$ the representation of any object

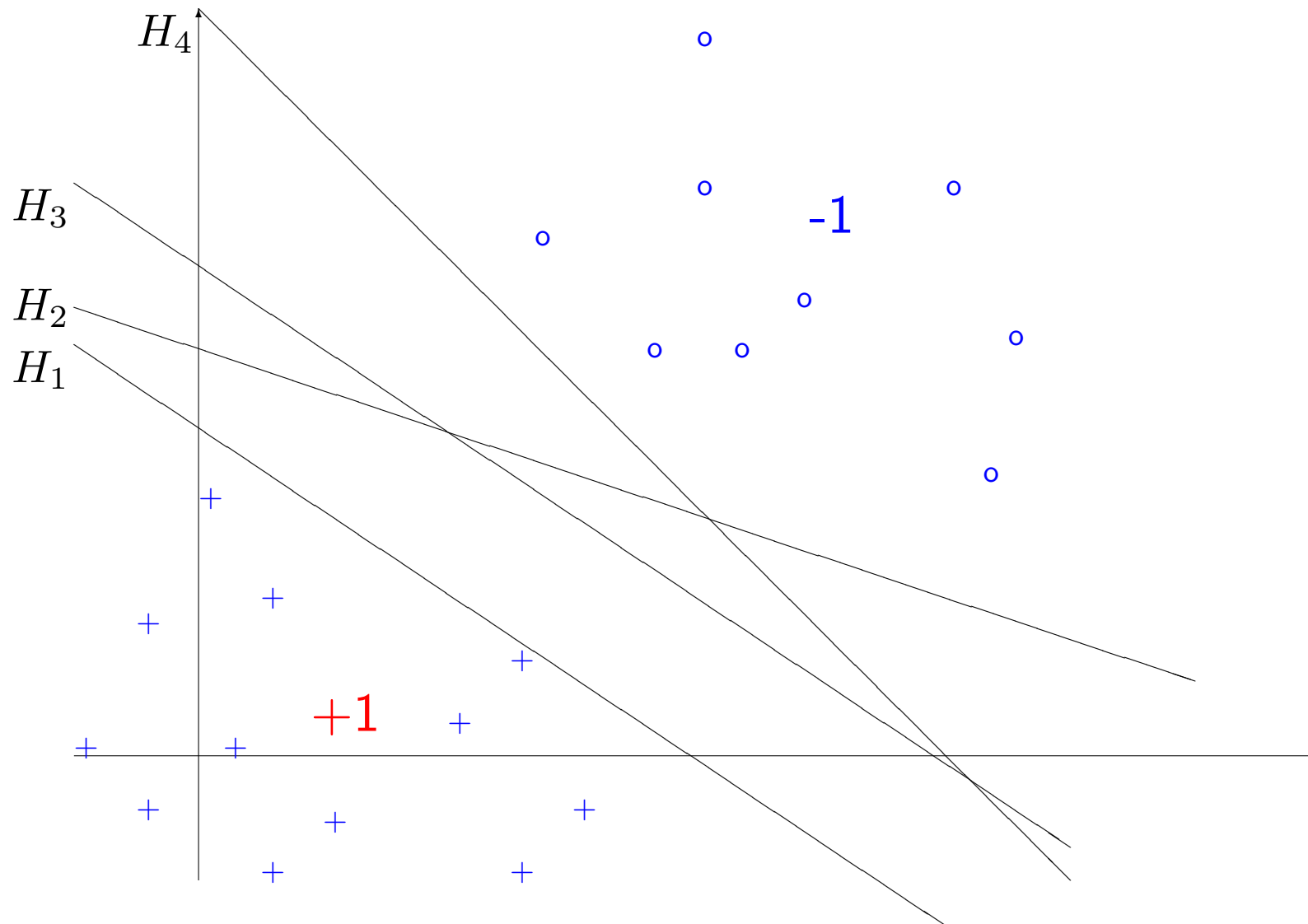
$$g_c(\mathbf{x}) = \sum_{j=1}^D w_{cj} \cdot x_j + w_{c0} = \mathbf{w}_c^t \mathbf{x} + w_{c0}, \quad 1 \leq c \leq C$$

- **Classification rule:** $\hat{c} = G(\mathbf{x}) \equiv \operatorname{argmax}_{1 \leq c \leq C} \mathbf{w}_c^t \mathbf{x} + w_{c0}$
- The *decision boundary* H_{ij} between any pair of classes i, j is:

$$H_{ij} = \{\mathbf{x} \in \mathbb{R}^D : \mathbf{w}_i^t \mathbf{x} + w_{i0} = \mathbf{w}_j^t \mathbf{x} + w_{j0}\}$$

- *Linear boundaries* or *hyperplanes* with dimension D (just lines if $D = 2$).

Linear discriminant functions



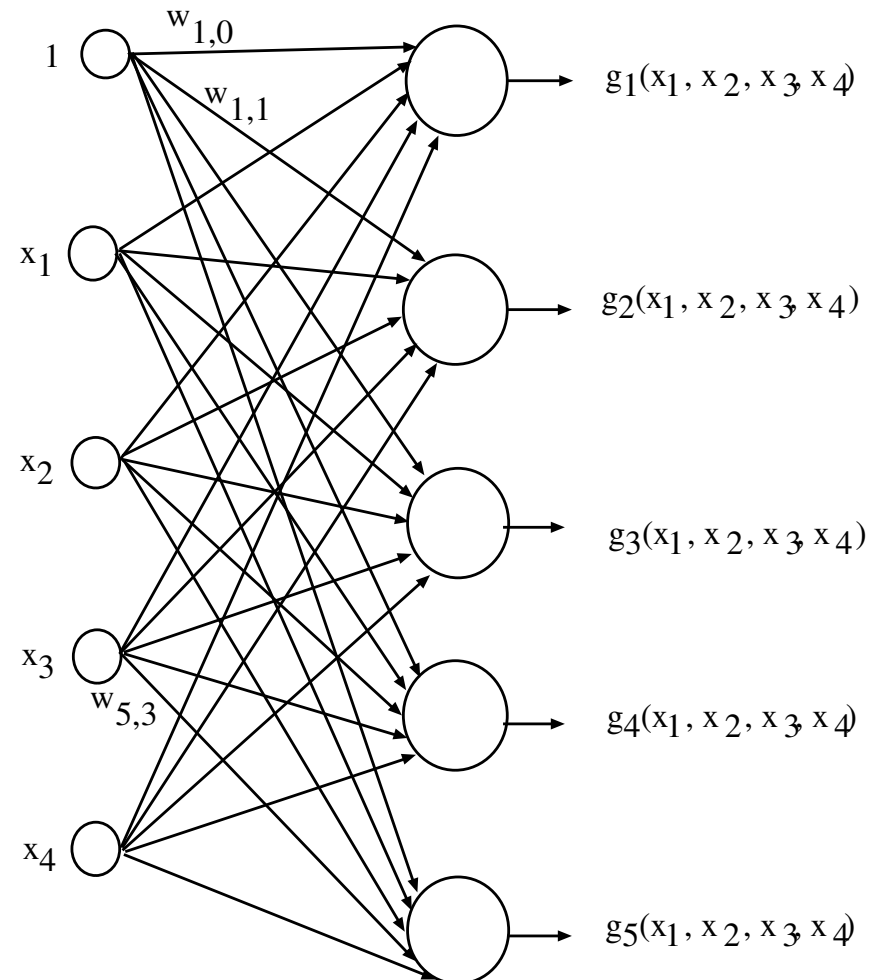
Linear Discriminant Functions

- Graphical representation of a LDF. Given an input $\mathbf{x} \in \mathbb{R}^D$

$$\hat{c} = G(\mathbf{x}) \equiv \operatorname{argmax}_{1 \leq c \leq C} g_c(\mathbf{x})$$

$$g_c(\mathbf{x}) = \mathbf{w}_c^t \mathbf{x} + w_{c0}$$

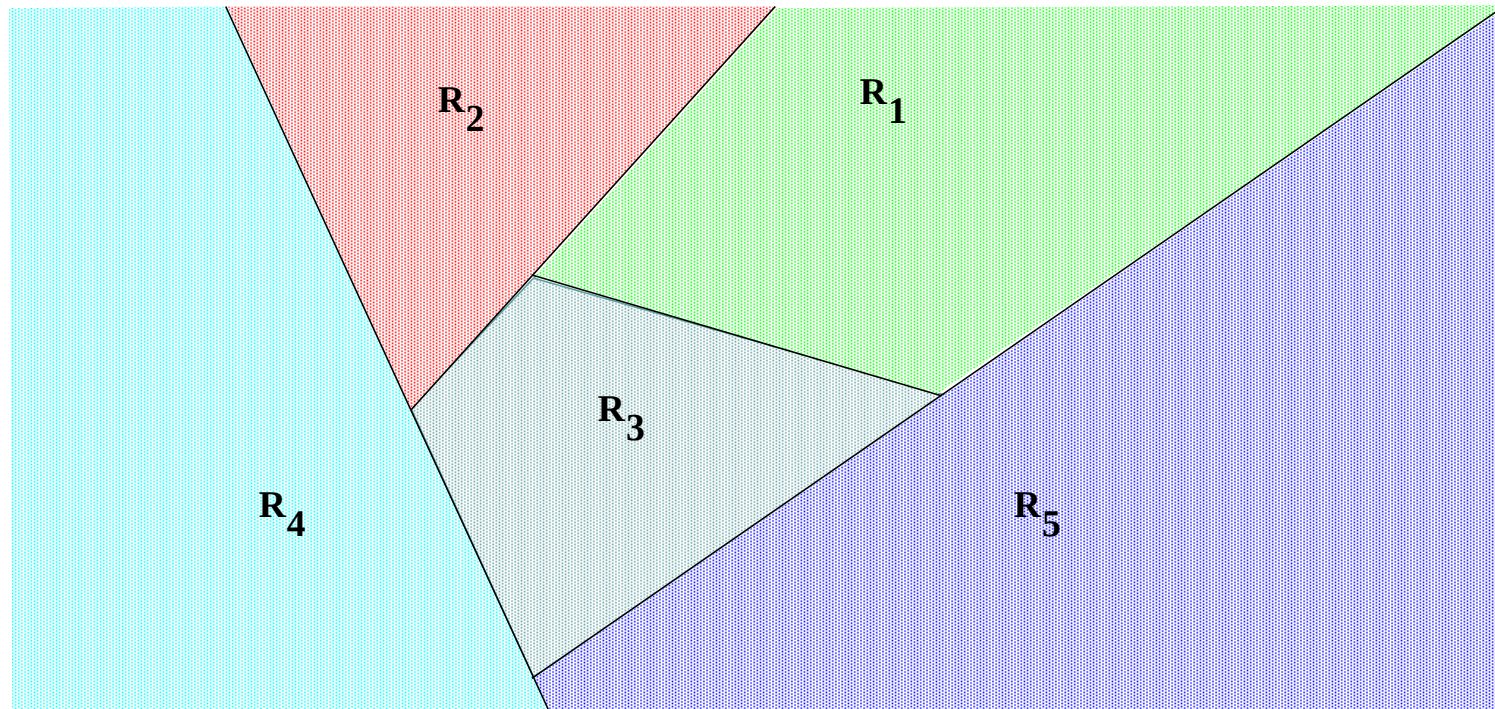
- $E = \mathbb{R}^4$,
- Classes = $\{1, 2, 3, 4, 5\}$



Decision surface of a LDF: Hyperplanes

$$H_{ij} = \{\mathbf{x} \mid g_i(\mathbf{x}) = g_j(\mathbf{x})\}$$

$$\mathbf{x} : \sum_{k=1}^D w_{ik} x_k + w_{i0} = \sum_{k=1}^D w_{jk} x_k + w_{j0} \Rightarrow \sum_{k=1}^D (w_{ik} - w_{jk}) x_k + (w_{i0} - w_{j0}) = 0$$



The problem of two classes

$$g_1, g_2 : \mathbb{R}^D \rightarrow \mathbb{R}$$

$$G(\mathbf{x}) = \begin{cases} 1 & \text{if } g_1(\mathbf{x}) > g_2(\mathbf{x}) \\ 2 & \text{if } g_1(\mathbf{x}) < g_2(\mathbf{x}) \end{cases}$$

Simplification:

$$g(\mathbf{x}) = g_1(\mathbf{x}) - g_2(\mathbf{x})$$

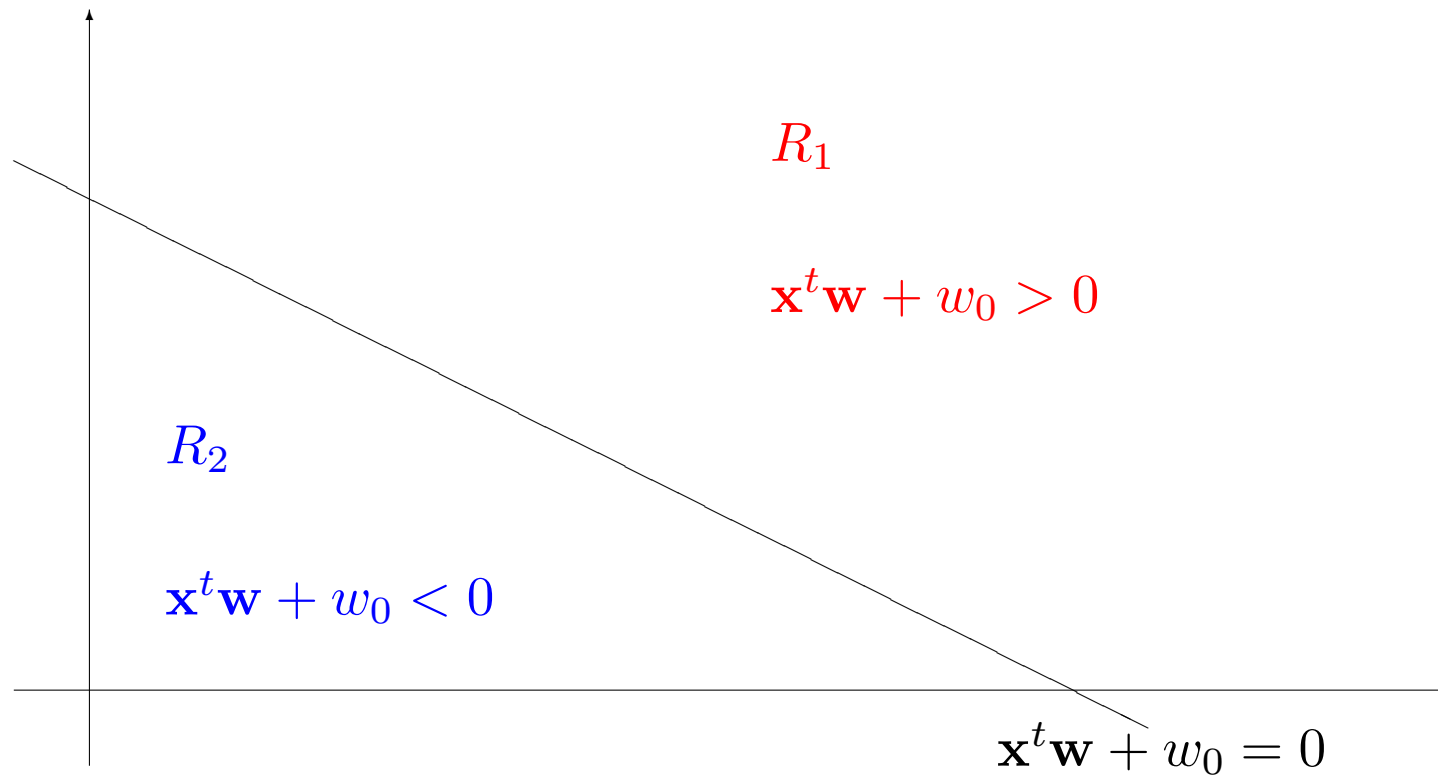
$$G(\mathbf{x}) = \begin{cases} 1 (+1) & \text{if } g(\mathbf{x}) > 0 \\ 2 (-1) & \text{if } g(\mathbf{x}) < 0 \end{cases}$$

$$g(\mathbf{x}) = \mathbf{x}^t (\mathbf{w}_1 - \mathbf{w}_2) + (w_{10} - w_{20}) = \mathbf{x}^t \mathbf{w} + w_0$$

Properties of the linear discriminant functions (2 classes)

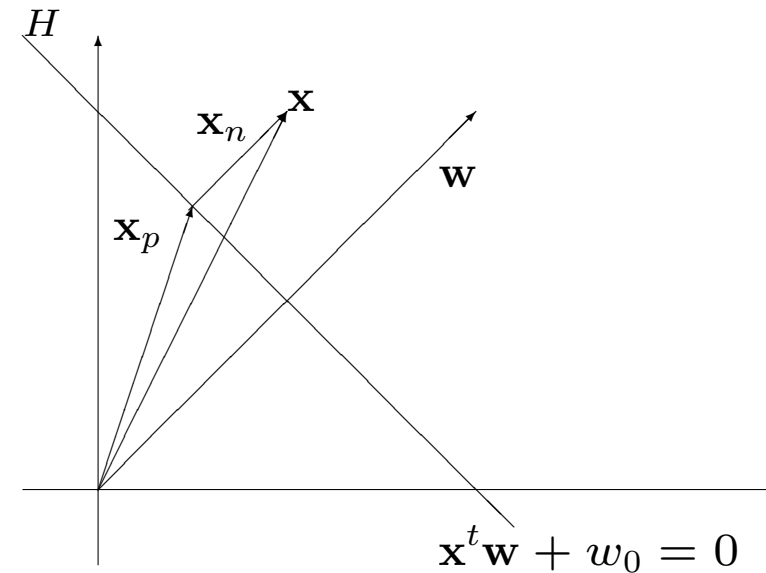
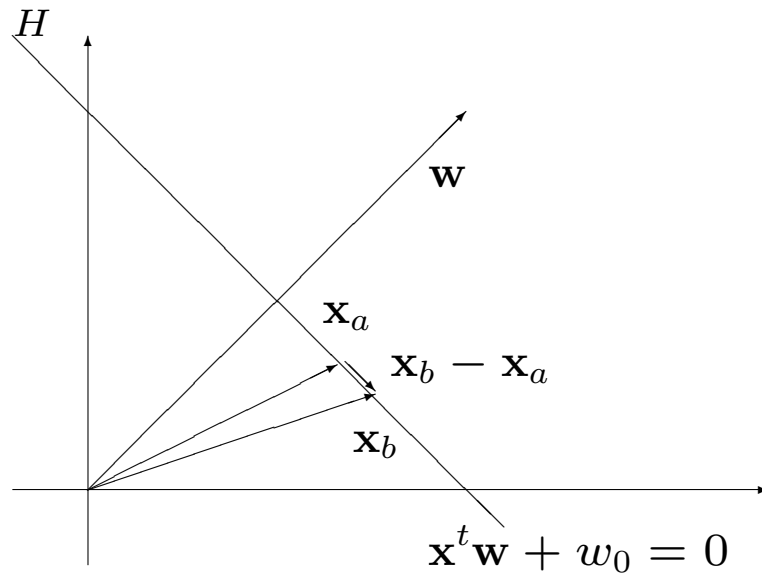
1. A LDF divides $\mathbb{R}^D$ in two semi-planes.
2. If $H = \{\mathbf{x} \mid g(\mathbf{x}) = 0\}$, H is orthogonal to $\mathbf{w}$.
3. If $w_0 > 0$, then the origin is in the positive part of H . If $w_0 < 0$, then the origin is in the negative part of H . If $w_0 = 0$, then H passes through the origin.
4. $r_{\mathbf{x}} = \frac{g(\mathbf{x})}{\|\mathbf{w}\|}$ is the distance of $\mathbf{x}$ to H .
5. The distance from the origin of coordinates to H is $w_0/\|\mathbf{w}\|$.
6. If $\gamma \in \mathbb{R}^+$, then $\gamma g(\cdot)$ and $g(\cdot)$ represents the same hyperplane of decision.

The problem of two classes



Properties of the linear discriminant functions

If $H = \{\mathbf{x} \mid g(\mathbf{x}) = 0\}$, H is orthogonal to $\mathbf{w}$ and $g(\mathbf{x}) = r_{\mathbf{x}} \|\mathbf{w}\|$, where $r_{\mathbf{x}}$ is the distance of $\mathbf{x}$ to H .



$$\begin{aligned} \mathbf{x}_a, \mathbf{x}_b \in H &\Rightarrow g(\mathbf{x}_a) = g(\mathbf{x}_b) \\ &\Rightarrow \mathbf{w}^t (\mathbf{x}_b - \mathbf{x}_a) = 0 \\ &\Rightarrow \mathbf{w} \perp H \end{aligned}$$

$$\begin{aligned} \mathbf{x} = \mathbf{x}_p + \mathbf{x}_n &\Rightarrow \mathbf{x} = \mathbf{x}_p + r \frac{\mathbf{w}}{\|\mathbf{w}\|} \\ &\Rightarrow g(\mathbf{x}) = w_0 + \mathbf{w}^t \mathbf{x} \\ &= w_0 + \mathbf{w}^t \mathbf{x}_p + r \frac{\|\mathbf{w}\|^2}{\|\mathbf{w}\|} \\ &= r \|\mathbf{w}\| \end{aligned}$$

Properties of the linear discriminant functions

If $\gamma \in \mathbb{R}^+$, then γg and g represents the same decision hyperplane.

- $\gamma \mathbf{w}$ and $\mathbf{w}$ is in the same direction
- $H_{\gamma g}$ and H_g are at the same distance of the origin of coordinates:

$$\frac{\gamma w_0}{\|\gamma \mathbf{w}\|} = \frac{w_0}{\|\mathbf{w}\|}$$

LDFs learning, problems of two classes

- Let $X = \{(\mathbf{x}_1, c_1), \dots, (\mathbf{x}_N, c_N)\}$, $\mathbf{x}_n \in \mathbb{R}^D$, $c_n \in \{+1, -1\}$ be a training set. X is **linear separable** (LS) if $\exists \mathbf{w} \in \mathbb{R}^D$ and $w_0 \in \mathbb{R}$ such that:

$$\forall n, 1 \leq n \leq N, \quad \mathbf{w}^t \mathbf{x}_n + w_0 \begin{cases} \geq 0 & \text{if } c_n = +1 \\ < 0 & \text{if } c_n = -1 \end{cases}; \quad \text{that is, } c_n (\mathbf{w}^t \mathbf{x}_n + w_0) \geq 0$$

- Learning: Given X , search for $\hat{\mathbf{w}}$ solves the system of N inequalities:

$$c_n (\mathbf{w}^t \mathbf{x}_n + w_0) \geq 0, \quad 1 \leq n \leq N$$

- Alternatively: Given X and a **margin** b , solves the system of N inequalities:

$$c_n (\mathbf{w}^t \mathbf{x}_n + w_0) \geq b, \quad 1 \leq n \leq N$$

- Equivalently: Given X and a **margin** b , search for $\hat{\mathbf{w}}$ minimizes:

$$q_X(\mathbf{w}, w_0) = \sum_{\substack{(\mathbf{x}, c) \in X \\ c (\mathbf{w}^t \mathbf{x} + w_0) < b}} -c (\mathbf{w}^t \mathbf{x} + w_0)$$

- Solution: Applying gradient descent to $q_X(\mathbf{w}, w_0)$

LDFs learning, *Perceptron* algorithm for problems of two classes

```
// Input:  $\mathbf{w}$  a vector with initial weights, and  $w_0$  an initial threshold
//          $(\mathbf{x}_1, c_1), \dots, (\mathbf{x}_N, c_N)$  with  $\mathbf{x}_n \in \mathbb{R}^D$ ,  $c_n \in \{-1, +1\}$  for  $1 \leq n \leq N$ ;
//          $\alpha \in \mathbb{R}^{>0}$ , the “learning factor”;
//          $b \in \mathbb{R}^{>0}$ , the “margin” (to control the convergence)
// Output:  $\mathbf{w}$  and  $w_0$  after the convergence
// Gradient descent on a function of the classification errors by  $\mathbf{w}$  and  $w_0$ 
repeat {
    error = 0;
    for all  $(n, 1 \leq n \leq N$  in a random way) {
         $g = c_n (\mathbf{w}^t \mathbf{x}_n + w_0)$ ;
        if  $(g < b)$  {
             $\mathbf{w} = \mathbf{w} + \alpha c_n \mathbf{x}_n$ ;  $w_0 = w_0 + \alpha c_n$ ;
            error ++;
        }
    }
} until (error = 0)
```

The final weights and threshold are: $\hat{\mathbf{w}} = \sum_{n=1}^N \beta_n c_n \mathbf{x}_n$ and $\hat{w}_0 = \sum_{n=1}^N \beta_n c_n$

LDFs learning, *Perceptron* algorithm for C classes

```
// Let be:  $\mathbf{w}_j$ ,  $1 \leq j \leq C$ ,  $C$  vectors with initial weights;  
//  $(\mathbf{x}_1, c_1), \dots, (\mathbf{x}_N, c_N)$ ,  $N$  learning samples;  
//  $\alpha \in \mathbb{R}^{>0}$ , the “learning factor”;  
//  $b \in \mathbb{R}$ , the “margin” (to control the convergence)
```

```
error = true
```

```
while (error) {
```

```
    for ( $n = 1$ ;  $n \leq N$ ;  $n++$ ) {
```

```
         $i = c_n$ ;  $g = \mathbf{w}_i^t \mathbf{x}_n + w_{i0}$ ; error=false
```

```
        for ( $j = 1$ ;  $j \leq C$ ,  $j \neq i$ ;  $j++$ ) {
```

```
            if ( $\mathbf{w}_j^t \mathbf{x}_n + w_{j0} + b > g$ ) {
```

```
                 $\mathbf{w}_j = \mathbf{w}_j - \alpha \mathbf{x}_n$ ;  $w_{j0} = w_{j0} - \alpha$ ; error=true
```

```
            }
```

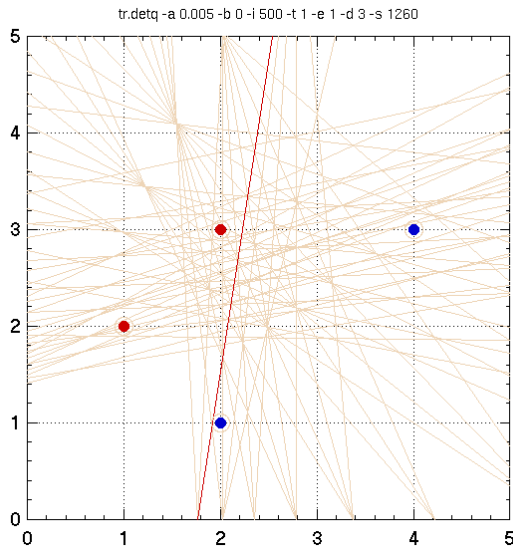
```
        if (error)  $\mathbf{w}_i = \mathbf{w}_i + \alpha \mathbf{x}_n$ ;  $w_{i0} = w_{i0} + \alpha$ ;
```

```
    }
```

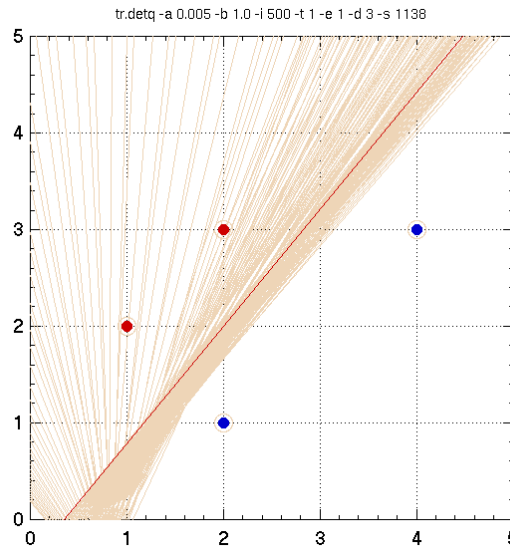
```
}
```

In case of an error the algorithm modifies the weights of the correct class and the weights of the incorrect class/es: $(\mathbf{w}_j^t \mathbf{x}_n + w_{j0} + b > g)$.

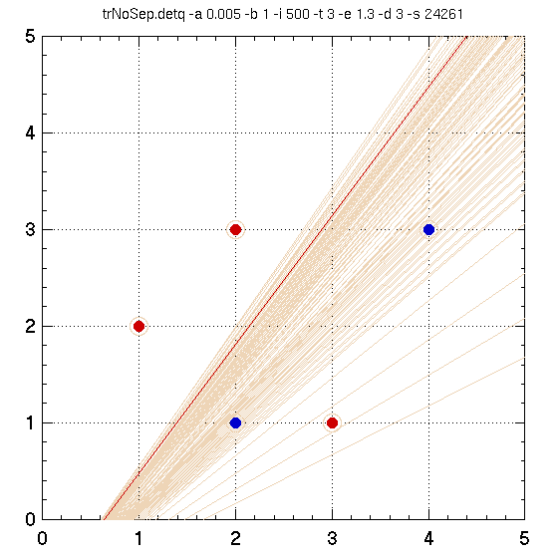
LDFs learning, *Perceptron* algorithm



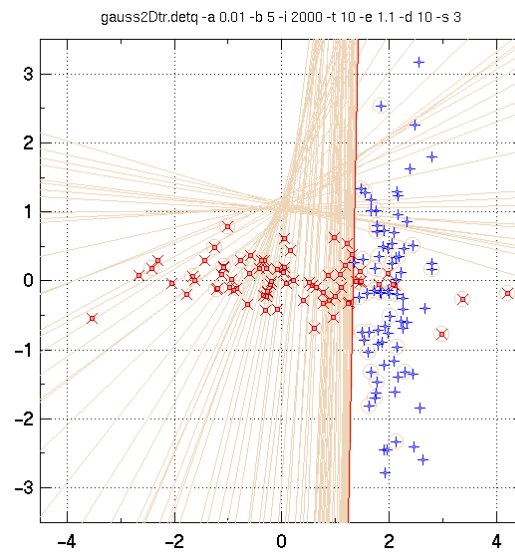
2 classes, margin 0



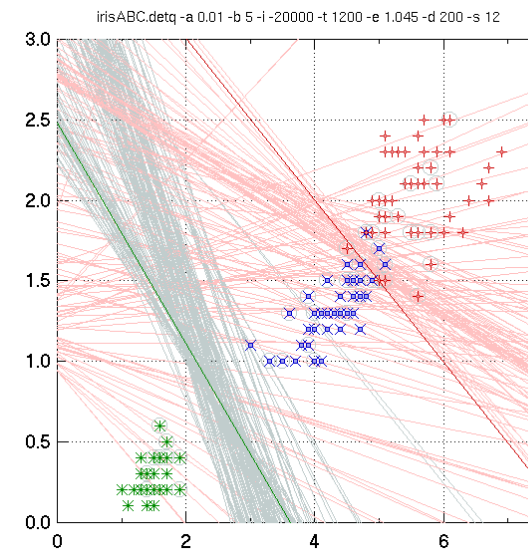
2 classes, margin > 0



2 classes, margin > 0 non-separable



2 gaussian classes



3 classes (iris)

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Introduction

- *Linear discriminant functions.* For $1 \leq c \leq C$:

$$g_c(\mathbf{x}) = \sum_{k=1}^D \mathbf{w}_{ck} x_k + \mathbf{w}_{c0}$$

- *Quadratic discriminant functions.* For $1 \leq c \leq C$:

$$g_c(\mathbf{x}) = \sum_{k_1=1}^D \sum_{k_2=1}^D \mathbf{w}_{ck_1 k_2} x_{k_1} x_{k_2} + \sum_{k=1}^D \mathbf{w}_{ck} x_k + \mathbf{w}_{c0}$$

- *Polynomial discriminant functions.* For $1 \leq c \leq C$:

$$g_c(\mathbf{x}) = \sum_{k_1, \dots, k_p} \mathbf{w}_{ck_1 \dots k_p} x_{k_1} \dots x_{k_p} + \dots + \sum_{k_1, k_2} \mathbf{w}_{ck_1 k_2} x_{k_1} x_{k_2} + \sum_{k=1}^D \mathbf{w}_{ck} x_k + \mathbf{w}_{c0}$$

Generalized linear discriminant functions

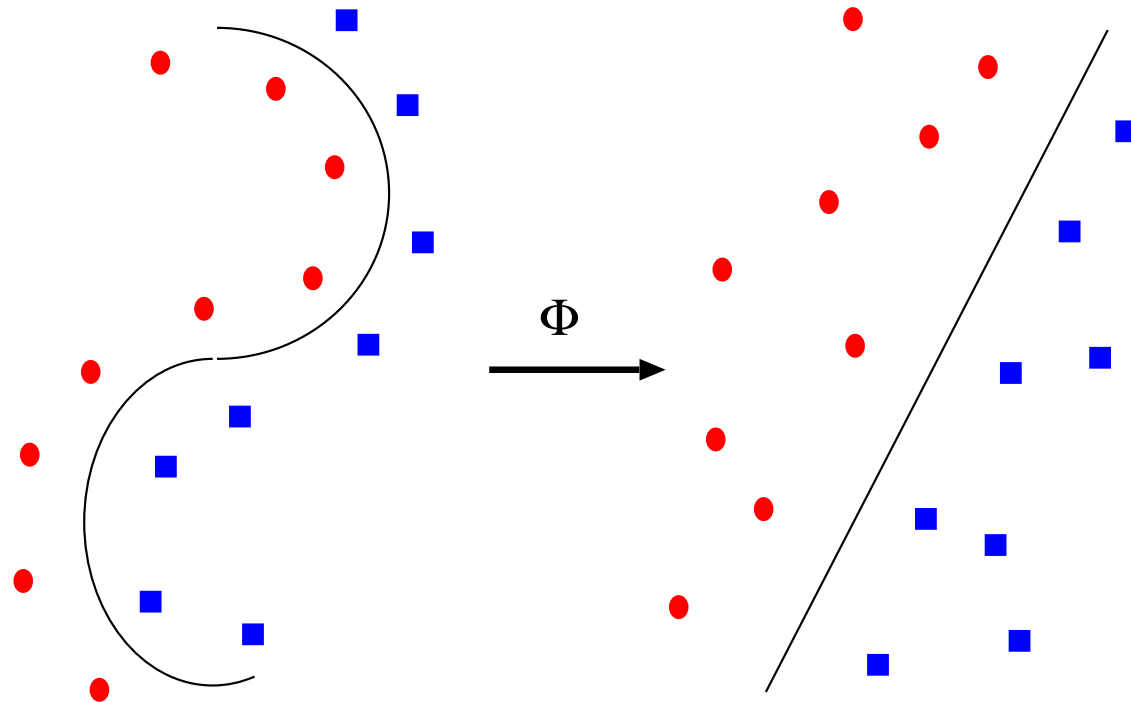
- *Generalized linear discriminant functions* (GLDF) For $1 \leq c \leq C$

$$g_c(\mathbf{x}) = \sum_{k=1}^{D'} w_{ck} \Phi_k(\mathbf{x}) + w_{c0} = \mathbf{w}_c^t \boldsymbol{\Phi}(\mathbf{x}) + w_{c0}$$

- For simplification, $\Phi_0(\mathbf{x}) = 1$, therefore $g_c(\mathbf{x}; \theta) = \sum_{k=0}^{D'} w_{ck} \Phi_k(\mathbf{x})$
- Feature space mapping through a non-linear function $\boldsymbol{\Phi} : \mathbb{R}^D \rightarrow \mathbb{R}^{\hat{D}}$ ($\hat{D} = D' + 1$)
- Typically $\hat{D} \geq D$: Dimensionality problem.

Generalized linear discriminant functions

Transformation from a non-linear separability to a linear separability



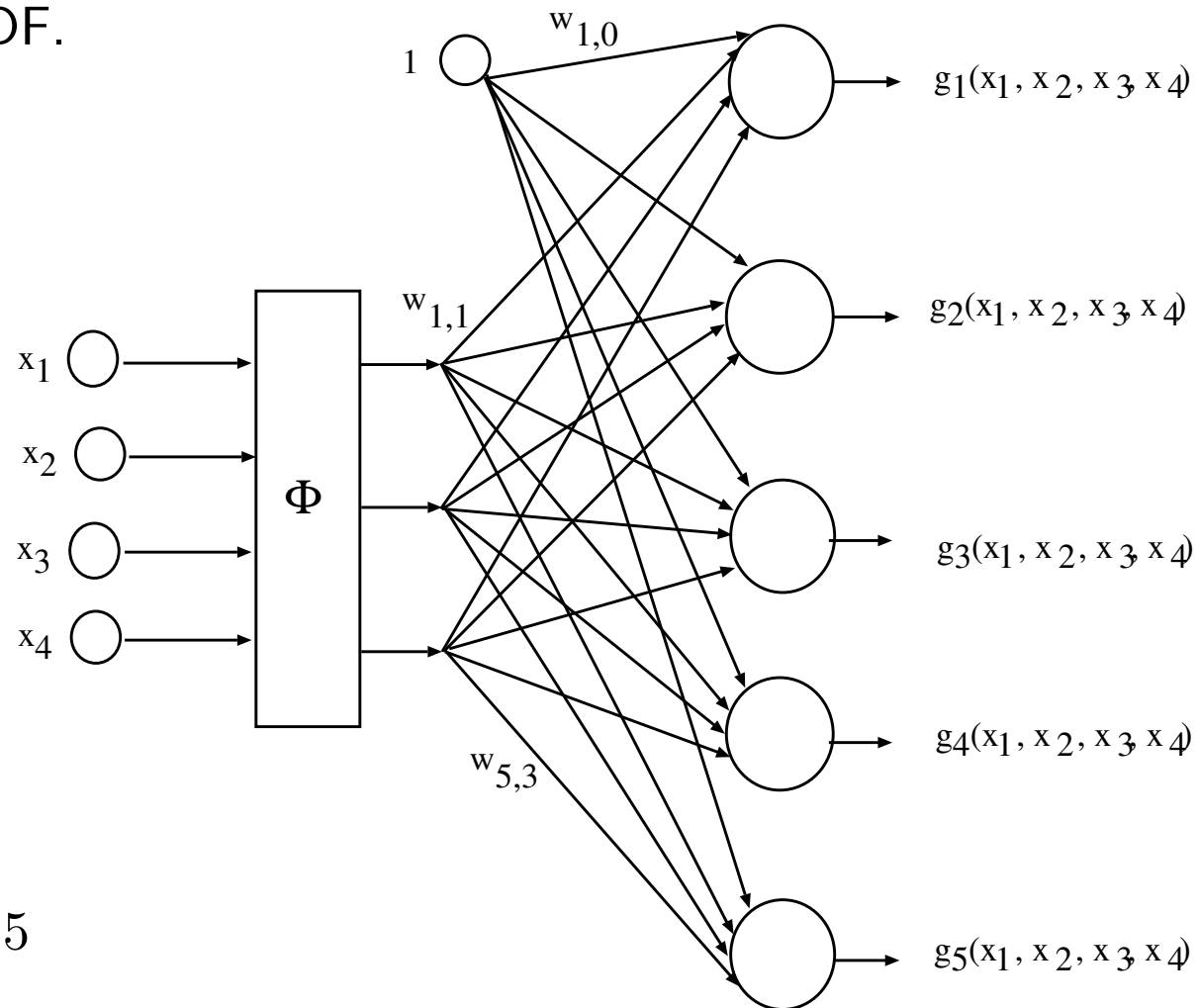
Generalized linear discriminant functions

- Graphical representation of a GLDF.
Given an input $\mathbf{x} \in \mathbb{R}^D$

$$\hat{c} = G(\mathbf{x}) \equiv \operatorname{argmax}_{1 \leq c \leq C} g_c(\mathbf{x})$$

$$g_c(\mathbf{x}) = \mathbf{w}_c^t \Phi(\mathbf{x}) + w_{c0}$$

- $E = \mathbb{R}^4$,
- $\Phi : E \rightarrow \mathbb{R}^3$
- Classes = $\{1, 2, 3, 4, 5\}$
- $\mathbf{w}_c \in \mathbb{R}^3$, $w_{c0} \in \mathbb{R}$ for $1 \leq c \leq 5$



Binary classification and kernels

- The Perceptron algorithm (and others) defines a LDF

$$g(\mathbf{x}) = \mathbf{w}^t \mathbf{x} + w_0 = \sum_{n=1}^N \beta_n c_n \mathbf{x}_n^t \mathbf{x} + \sum_{n=1}^N \beta_n c_n$$

- If the training set is not linearly separable, we need to change the representation space to obtain linear separability.
- We obtain an implicit change of the representation space using the *kernel trick*:

$$g(\mathbf{x}) = \sum_{n=1}^N \beta_n c_n \Phi(\mathbf{x}_n)^t \Phi(\mathbf{x}) + \sum_{n=1}^N \beta_n c_n = \sum_{n=1}^N \beta_n c_n K(\mathbf{x}_n, \mathbf{x}) + \sum_{n=1}^N \beta_n c_n$$

Binary classification and Kernels

- We define a kernel function as:

$$K : \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$$

Such that:

$$K(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x})^t \Phi(\mathbf{y})$$

- But this alternative representation Φ is not performed, we just need to know the scalar product on this new representation space.

Kernels: an example

Let $\mathbf{x} = (x_1, x_2, x_3)^t$, $\mathbf{y} = (y_1, y_2, y_3)^t$ and $K: \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ defined as:

$$K(\mathbf{x}, \mathbf{y}) = (x_1 y_1 + x_2 y_2 + x_3 y_3)^2$$

Is $K(\mathbf{x}, \mathbf{y})$ a kernel? . . . Yes:

$$\begin{aligned} K(\mathbf{x}, \mathbf{y}) &= (x_1 y_1 + x_2 y_2 + x_3 y_3)^2 \\ &= x_1^2 y_1^2 + x_2^2 y_2^2 + x_3^2 y_3^2 + 2 x_1 y_1 x_2 y_2 + 2 x_1 y_1 x_3 y_3 + 2 x_2 y_2 x_3 y_3 \end{aligned}$$

$K(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x})^t \Phi(\mathbf{y})$ si $\Phi: \mathbb{R}^3 \rightarrow \mathbb{R}^6$ is defined as:

$$\Phi(\mathbf{x}) = (x_1^2, x_2^2, x_3^2, \sqrt{2} x_1 x_2, \sqrt{2} x_1 x_3, \sqrt{2} x_2 x_3)^t$$

$$\Phi(\mathbf{y}) = (y_1^2, y_2^2, y_3^2, \sqrt{2} y_1 y_2, \sqrt{2} y_1 y_3, \sqrt{2} y_2 y_3)^t$$

Alternatives for computing $K(\mathbf{x}, \mathbf{y})$:

- *Direct* in $\mathbb{R}^3$, using $K(\mathbf{x}, \mathbf{y})$: $3 + 2 + 1 = 6$ products + additions
- Compute first $\Phi(\mathbf{x})$, $\Phi(\mathbf{y})$ in $\mathbb{R}^6$ and compute next $\Phi(\mathbf{x})^t \Phi(\mathbf{y})$:
 $2 \cdot 6 + 6 + 5 = 23$ products + additions

Learning - Kernel Perceptron

- The Kernel Perceptron algorithm learns the following function:

$$g(\mathbf{x}) = \sum_{n=1}^N \beta_n c_n K(\mathbf{x}_n, \mathbf{x}) + \sum_{n=1}^N \beta_n c_n$$

- The parameters to learn are β_n for $1 \leq n \leq N$
- In this learning stage the kernel function is just a matrix K with all the pairs $K(\mathbf{x}_i, \mathbf{x}_j)$

Learning - Kernel Perceptron

// Input: $(\mathbf{x}_1, c_1), \dots, (\mathbf{x}_N, c_N)$ with $\mathbf{x}_n \in \mathbb{R}^D$, $c_n \in \{-1, +1\}$ for $1 \leq n \leq N$;
// Output: β_n for $1 \leq n \leq N$

$\beta_n = 0$ for $1 \leq n \leq N$; ;

repeat

error = 0;

for all $(n, 1 \leq n \leq N \text{ in a random way})$ {

$g = c_n \left(\sum_{n'=1}^N \beta_{n'} c_{n'} K(\mathbf{x}_{n'}, \mathbf{x}_n) + \sum_{n'=1}^N \beta_{n'} c_{n'} \right);$

if $(g < b)$ {

$\beta_n ++;$

error ++;

}

}

} until (error = 0)

Exercise

- Given the following Kernel matrix:

$$K = \begin{bmatrix} 1 & 1/3 & 1/3 & 1/3 & 1/5 \\ 1/3 & 1 & 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1 & 1/5 & 1/3 \\ 1/3 & 1/3 & 1/5 & 1 & 1/3 \\ 1/5 & 1/3 & 1/3 & 1/3 & 1 \end{bmatrix}$$

where $X = \{(\mathbf{x}_1, +1), (\mathbf{x}_2, -1), (\mathbf{x}_3, +1), (\mathbf{x}_4, -1), (\mathbf{x}_5, +1)\}$

- Obtain the β_i with the Kernel Perceptron algorithm and $b = 1$

Most used Kernels

- A polynomial kernel is: $K(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \cdot \mathbf{y} + c)^d$
- The gaussian kernel is: $K(\mathbf{x}, \mathbf{y}) = \exp\left(\frac{-\|\mathbf{x}-\mathbf{y}\|^2}{2\sigma^2}\right)$

In an infinity dimension space the training samples are linearly separables

Generalized kernels

- How to obtain new kernels functions
- It is necessary to demonstrate that:

$$\exists \Phi : \mathbb{R}^D \rightarrow \mathbb{R}^{D'} : K(\mathbf{y}, \mathbf{x}) = \Phi(\mathbf{y})\Phi(\mathbf{x})$$

- ***Mercer condition***: a necessary and sufficient condition for K to be a valid kernel is that the Gram matrix: $K(\mathbf{x}_i, \mathbf{x}_j)$ has to be positive-semidefinite for all possible pair of the training set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$.
- A matrix K is positive-semidefinite if:

$$\mathbf{x}^t K \mathbf{x} \geq 0, \forall \mathbf{x} \in \mathbb{R}^D$$

Generalized Kernels

If K_1 and K_2 are kernels, then K is a kernel:

$$K(\mathbf{x}, \mathbf{y}) = \left\{ \begin{array}{ll} c \cdot K_1(\mathbf{x}, \mathbf{y}) & c > 0 \\ f(\mathbf{x}) \cdot K_1(\mathbf{x}, \mathbf{y}) \cdot f(\mathbf{y}) & \text{for any function } f \\ q(K_1(\mathbf{x}, \mathbf{y})) & q \text{ a polynomial with non-negative coefficients} \\ (c + K_1(\mathbf{x}, \mathbf{y}))^d & d, c > 0 \\ K_1(\mathbf{x}, \mathbf{y}) + K_2(\mathbf{x}, \mathbf{y}) \\ K_1(\mathbf{x}, \mathbf{y}) \times K_2(\mathbf{x}, \mathbf{y}) \\ \exp(K_1(\mathbf{x}, \mathbf{y})) \\ \frac{K_1(\mathbf{x}, \mathbf{y})}{\sqrt{K_2(\mathbf{x}, \mathbf{y})}} \end{array} \right.$$

Exercise

Given the following kernel function: $K(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \cdot \mathbf{y} + 1)^2$ and the following training set $X = \{(\mathbf{x}_1, +1), (\mathbf{x}_2, +1), (\mathbf{x}_3, +1), (\mathbf{x}_4, -1), (\mathbf{x}_5, -1), (\mathbf{x}_6, -1)\}$ with:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \mathbf{x}_5 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{x}_6 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- a) Compute the Kernel matrix K from the training set
- b) Perform one iteration (from $\mathbf{x}_1$ to $\mathbf{x}_6$) of the Kernel Perceptron algorithm
- c) Clasify the new test sample $\mathbf{x} = (0, 2)^t$ with the weights β obtained in the previous step

Radial basis function networks

- Given a vector μ , a **radial basis function** (RBF) is $\phi_{\mu}(\mathbf{x}) = \phi(\|\mathbf{x} - \mu\|)$

Some type of function:
$$\left\{ \begin{array}{l} - \text{Gaussian function: } \phi(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right) \\ - \text{Quadratic function: } \phi(r) = (r^2 + \sigma^2)^{1/2} \\ - \text{"Thin-plane-spline" function: } \phi(r) = r^2 \log r \end{array} \right.$$

- RBFs allow to define GLDF as: Given M vectors $\mu_m \in \mathbb{R}^D$, $1 \leq m \leq M$

$$g_k(\mathbf{x}; \theta) = \sum_{m=1}^M w_{k,m} \phi(\|\mathbf{x} - \mu_m\|) = \mathbf{w}_k^t \Phi(\mathbf{x}) \quad 1 \leq k \leq C$$

- Problem:** Given $A = \{(\mathbf{x}_n, \mathbf{t}_n) \mid \mathbf{x}_n \in \mathbb{R}^D, \mathbf{t}_n \in \mathbb{R}^C\}_{1 \leq n \leq N}$ search for vectors $\mathbf{w}_c \in \mathbb{R}^N$ ($1 \leq c \leq C$) and vectors $\mu_k \in \mathbb{R}^d$ ($1 \leq k \leq M$) such that $g(\mathbf{x}_n; \theta) = \mathbf{t}_n$ for $1 \leq n \leq N$.

Learning with radial basis function networks

Given $A = \{(\mathbf{x}_1, \mathbf{t}_1), \dots, (\mathbf{x}_n, \mathbf{t}_n)\}$, with $\mathbf{x}_i \in \mathbb{R}^D$, $\mathbf{t}_i \in \mathbb{R}^C$,

- **Sequential learning** of the radial basis functions (μ_k and σ_k) and the weight ($\mathbf{w}_y$):
 1. Learning radial basis functions ϕ from $A' = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$,
 - using *clustering*
 - using *mixtures of gaussians*
 - using *autoorganizative maps*
 2. Learning weights $\mathbf{w}$ from $A'' = \{(\phi(\mathbf{x}_1), \mathbf{t}_1), \dots, (\phi(\mathbf{x}_N), \mathbf{t}_N)\}$,
 - Perceptron
 - Pocket Perceptron
 - Widrow-Hoff
- **Integrate learning** of radial basis function and the weights by minimizing the mean squared error.

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Introduction

- Motivation: find linear classifiers with good generalization
- Lagrange optimization problem
- Convex optimization problem
- Extension to kernel to deal with non-linear boundaries
- Soft-constraint to avoid complex boundaries

Maximum margin classifiers

A *canonical LDF* w.r.t. a set X of N samples is $\mathbf{w} \equiv (\mathbf{w}, w_0)$, such that

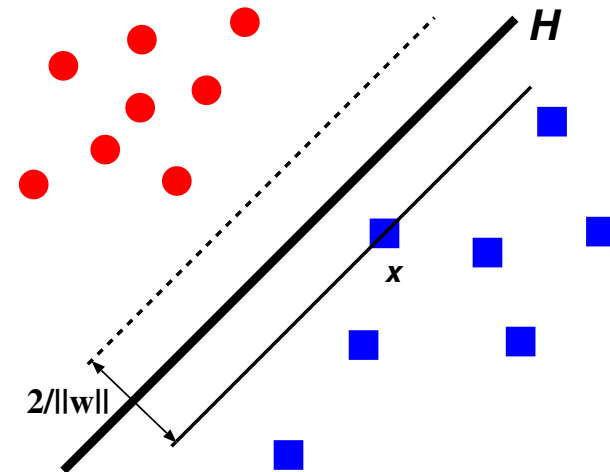
$$\min_{1 \leq n \leq N} |g(\mathbf{x}_n; \mathbf{w})| = \min_{1 \leq n \leq N} |\mathbf{w}^t \mathbf{x}_n + w_0| = 1$$

The distance r of the nearest $\mathbf{x} \in X$ to H is:

$$r = \frac{|\mathbf{w}^t \mathbf{x} + w_0|}{\|\mathbf{w}\|} = \frac{1}{\|\mathbf{w}\|}$$

And the *margin* of H with respect to X is:

$$2r = \frac{2}{\|\mathbf{w}\|}$$



Given a (linearly separable) training data set: $X = \{(\mathbf{x}_1, c_1), \dots, (\mathbf{x}_N, c_N)\}$, $\mathbf{x}_i \in \mathbb{R}^d$ and $c_i \in \{-1, +1\}$, find a canonical LDF w.r.t to X that classify correctly all the samples with maximum margin:

- **maximize** $\frac{2}{\|\mathbf{w}\|}$
- **subject to** $c_i(\mathbf{w}\mathbf{x}_i + w_0) \geq 1$, for all $1 \leq i \leq N$

Support Vector Machines. Linear SVM

- Alternatively, we can formulate the previous optimization problem as: Given a LS set X of N samples:
 - minimize $\frac{1}{2}\|\mathbf{w}\|^2$
 - subject to $c_i(\mathbf{w}\mathbf{x}_i + w_0) \geq 1$, for all $1 \leq i \leq N$
- By using the Lagrange multipliers method, it is necessary to solve the Dual Lagrange optimization problem (α_i for $1 \leq i \leq N$ are known as Lagrange multipliers):
 - maximize $\Lambda_D(\boldsymbol{\alpha}) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j c_i c_j \mathbf{x}_i^T \mathbf{x}_j$
 - subject to $\sum_{i=1}^N \alpha_i c_i = 0$ and $\alpha_i \geq 0$ for $1 \leq i \leq N$

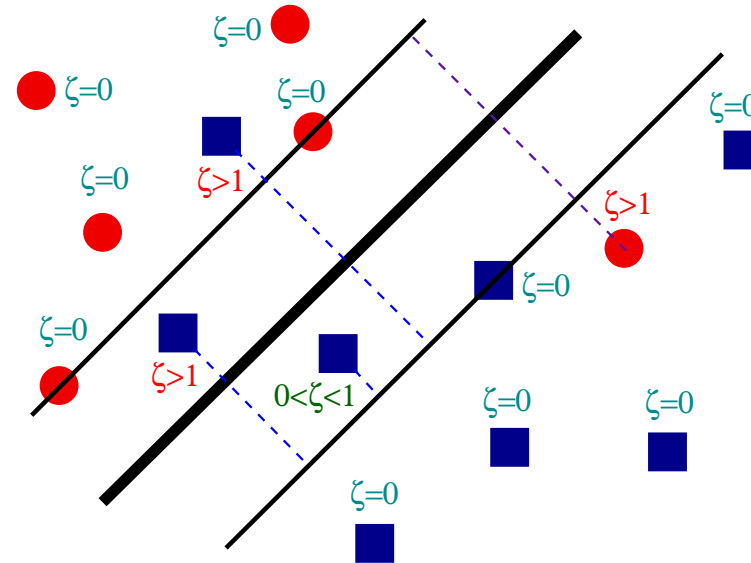
This problem can now be solved by standard quadratic programming techniques.

- Support vectors $S \subset X$: The subset of samples $(\mathbf{x}_n, c_n) \in X$ s.t. $\alpha_n^* \neq 0$
- The solution is $g_c(\mathbf{x}) = \sum_{(\mathbf{x}_n, c_n) \in S} \alpha_n^* c_n \mathbf{x}_n^t \mathbf{x} + w_0$ with $w_0 = c_n - \mathbf{w}^{*t} \mathbf{x}_n$, for a n s.t. $0 < \alpha_n^*$

Support Vector Machines. Soft Margin

- Idea: to allow mislabeled examples
- Slack variables:

The method introduces non-negative slack variables, ξ_i , which measure the degree of misclassification of the data x_i : $c_i(\mathbf{w}\mathbf{x}_i + w_0) \geq 1 - \xi_i$



- The objective function is then increased by a function which penalizes non-zero ξ_i :

- minimize $\frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i$
- subject to $c_i(\mathbf{w}\mathbf{x}_i + w_0) \geq 1 - \xi_i$ and $\xi_i \geq 0$, for all $1 \leq i \leq N$

Support Vector Machines. Soft Margin

- By using the Lagrange multipliers method, it is necessary to solve the Dual Lagrange optimization problem (α_i for $1 \leq i \leq N$ are known as Lagrange multipliers):

- maximize $\Lambda_D(\boldsymbol{\alpha}) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j c_i c_j \mathbf{x}_i^T \mathbf{x}_j$

- subject to $\sum_{i=1}^N \alpha_i c_i = 0$ and $C \geq \alpha_i \geq 0$ for $1 \leq i \leq N$

This problem can now be solved by standard quadratic programming techniques.

- Support vectors $S \subset X$: The subset of samples $(\mathbf{x}_n, c_n) \in X$ s.t. $\alpha_n^* \neq 0$
- The solution is $g_c(\mathbf{x}) = \sum_{(\mathbf{x}_n, c_n) \in S} \alpha_n^* c_n \mathbf{x}_n^t \mathbf{x} + w_0$ with $w_0 = c_n - \mathbf{w}^{*t} \mathbf{x}_n$, for a n s.t. $0 < \alpha_n^* < C$

Exercise

- Given a training set:

$$S = \{((1, 4), +1), ((2, 2), +1), ((2, 3), +1), ((4, 2), +1), ((3, 4), -1), ((3, 5), -1), ((5, 4), -1), ((5, 6), -1), ((4, 4), +1), ((4, 3), -1)\},$$

the optimal Lagrange multipliers α_n^* with $C = 1000$ obtained are:

$$[250.87, 0.0, 0.0, 500.75, 751.62, 0.0, 0.0, 0.0, 1000.0, 1000.0]$$

Write the corresponding linear discriminant function, the slack variables ξ_n^* and classify the sample $(4, 5)$.

Support Vector Machines. Kernel SVM

- Kernel extension:

Maximize,

$$\Lambda_D(\boldsymbol{\alpha}) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j c_i c_j K(\mathbf{x}_i, \mathbf{x}_j) \text{ s.t. } \sum_{i=1}^n \alpha_i c_i = 0 \text{ and } \alpha_i \geq 0$$

where $K(\cdot, \cdot)$ is a kernel function

- The solution is

$$g_c(\mathbf{x}) = \sum_{(\mathbf{x}_n, c_n) \in S} \alpha_n^* c_n K(\mathbf{x}_n, \mathbf{x}) + w_0$$

with $w_0 = c_n - \sum_{m=1}^N \alpha_m^* c_m K(\mathbf{x}_m, \mathbf{x}_n)$: for a n s.t. $0 < \alpha_n^* < C$

- The parameter selection of C and the Kernel function is carried out by cross-validation

Support Vector Machines for C classes

- **One-against-one plus DAG**: $C(C - 1)/2$ classifiers; classification directed by a DAG (direct acyclic graph)
- **One-against-rest**: C discriminant functions.
- **SVM^{multiC}**: Direct optimization in C classes [Cramer & Singer, 01]
- **Kesler construction**: Transform a problem of C classes into another of 2 classes (with high dimension). [Duda & Hart, 73], [Franc & Hlaváč, 02]

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